PRACTICAL NO. 1

Introduction to DevOps

LOB1 To Learn what DevOps is, including its principles and the benefits it offers to organizations.

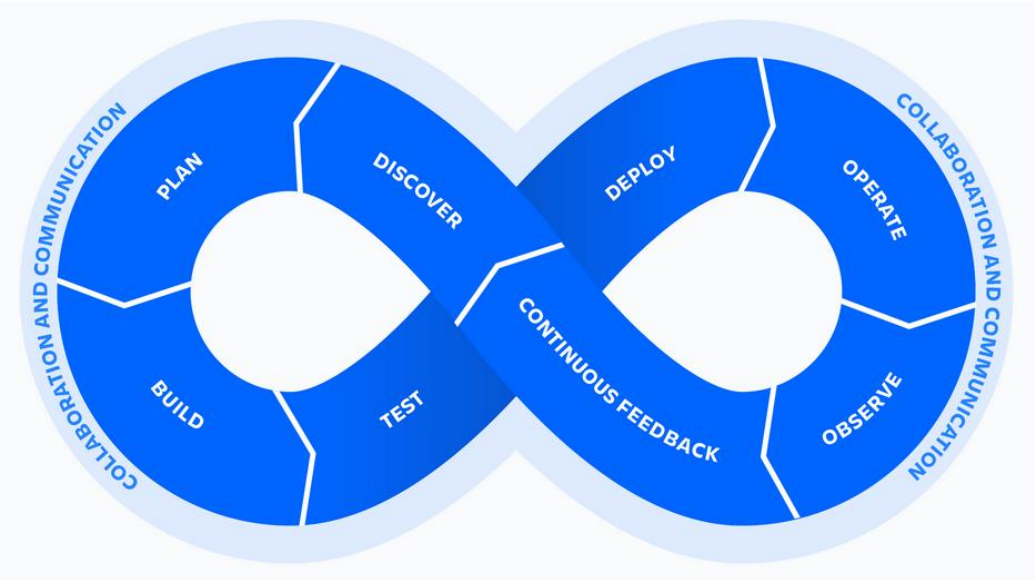
LO1 Recall and explain the key principles and benefits of DevOps.

1. What is DevOps?

DevOps is the combination of cultural philosophies, practices, and tools to enhance an organization's capability to deliver applications and services rapidly. By fostering faster product evolution and improvement compared to traditional software development and infrastructure management methods, DevOps empowers organizations to better meet customer needs and maintain a competitive edge in the market.

DevOps is a methodology that combines and automates software development (Dev) and information technology operations (Ops) to:

- Improve and shorten the systems development life cycle.
- Collaborate between developers and operations engineers.
- Create a culture where everyone works toward the same goal.



2. Key Principles of DevOps

DevOps is guided by a set of principles that promote collaboration, efficiency, and quality in software development and IT operations. These principles form the foundation for DevOps practices.

a. Collaboration and Communication

i. Break down silos between development, operations, and other teams.
ii. Foster open communication and shared goals.
iii. Use collaboration tools like Slack, Microsoft Teams, or Jira to streamline workflows.

Example: Developers and operations teams jointly plan deployment strategies to avoid downtime during a product launch.

b. Continuous Integration (CI)

i. Developers frequently merge code changes into a shared repository.
ii. Automated tests run to identify integration issues early.

Example: A team uses Jenkins or GitLab CI to automatically test and integrate code after each commit.

c. Continuous Delivery (CD)

i. Ensure that code changes are automatically prepared for deployment.
ii. Focus on frequent, reliable releases to production.

Example: An e-commerce company uses AWS CodePipeline to deploy updates daily without affecting user experience.

d. Automation

i. Automate repetitive tasks like testing, building, and deploying software.
ii. Use tools for infrastructure provisioning, monitoring, and scaling.

Example: A team uses Ansible to automate server configuration and deployment, saving hours of manual work.

e. Infrastructure as Code (IaC)

i. Manage infrastructure using code, ensuring consistency and scalability.
ii. Tools like Terraform and AWS CloudFormation help define infrastructure in code.

Example: A company uses IaC to replicate its production environment for testing, reducing setup errors.

f. Monitoring and Feedback

i. Continuously monitor applications and infrastructure for performance and issues.
ii. Use feedback loops to improve processes and resolve problems quickly.

Example: Netflix uses Prometheus and Grafana to monitor server health and optimize performance.

g. Continuous Testing

i. Test software automatically at every stage of the development lifecycle.
ii. Ensure quality and security before deployment.

Example: A team integrates Selenium for automated UI testing in its CI/CD pipeline.

h. Shared Responsibility

i. Developers and operations teams share accountability for the success of applications.
ii. Teams collaborate to resolve issues and improve the product.

Example: Both developers and operations participate in post-mortem analyses after a production outage.

3. Key Practices of DevOps

1. Version Control: Use tools like Git to manage code changes and enable collaboration.
2. CI/CD Pipelines: Automate building, testing, and deploying code.
3. Microservices Architecture: Break applications into smaller, independent services.
4. Containerization: Use Docker or Kubernetes to ensure consistency across environments.
5. Agile Development: Adopt Agile methodologies to enhance flexibility and adaptability.
6. Incident Management: Implement robust systems for detecting, reporting, and resolving incidents.
7. Change Management: Manage changes effectively to minimize risks and downtime.

By adopting these principles and practices, organizations can achieve faster software delivery, higher quality, and improved collaboration across teams.

4. Benefits of Implementing DevOps

a. Faster Time to Market - Accelerates the development and deployment process, enabling organizations to deliver new features, updates, and bug fixes faster.
b. Improved Collaboration and Efficiency - Breaks down silos between development and operations teams, fostering better communication and shared responsibilities.
c. Reduced Costs - Automation and streamlined workflows minimize manual effort, reducing operational costs.
d. Continuous Improvement - Feedback loops and monitoring allow organizations to learn from successes and failures, improving processes over time.
e. Increased Deployment Frequency - DevOps enables frequent, incremental updates instead of infrequent, large releases.
f. Enhanced Customer Satisfaction - Faster delivery of new features and higher-quality applications result in better user experiences and happier customers.
g. Scalability and Flexibility - Infrastructure as Code (IaC) and containerization allow applications to scale easily with demand.
h. Better Incident Management - Continuous monitoring and quick feedback loops ensure faster identification and resolution of issues.

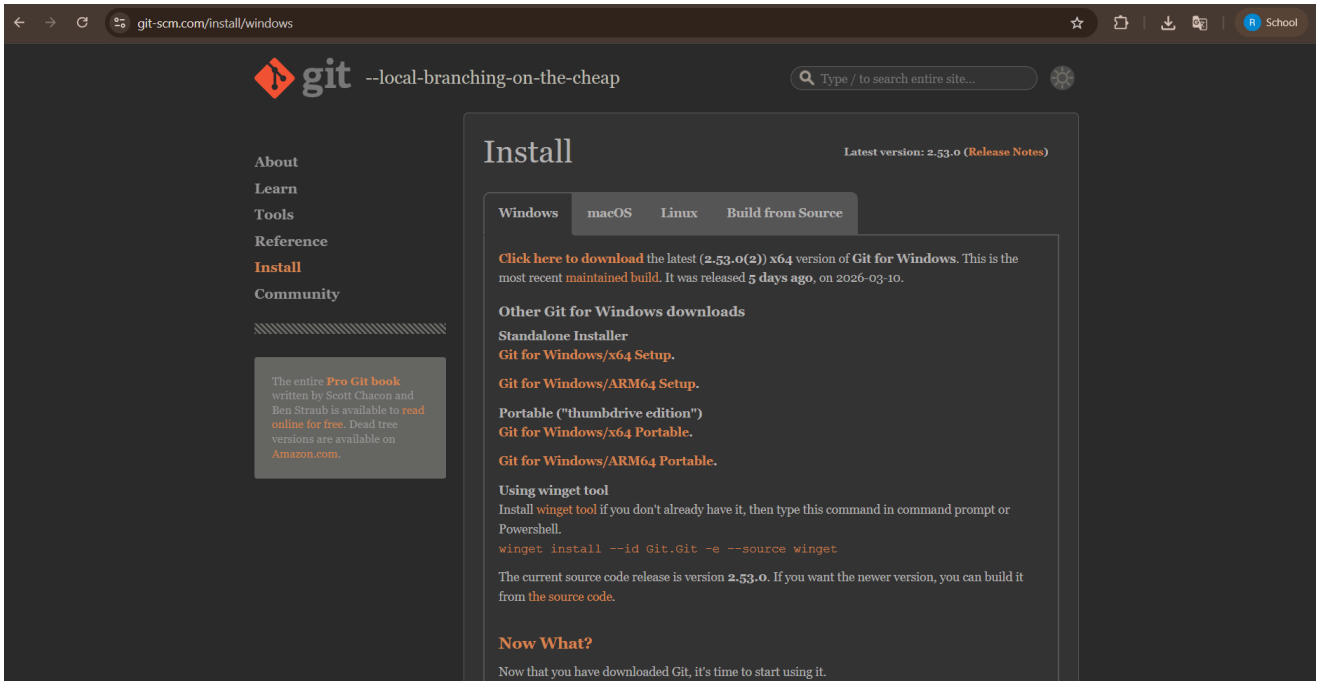
5. Basic Git Setup

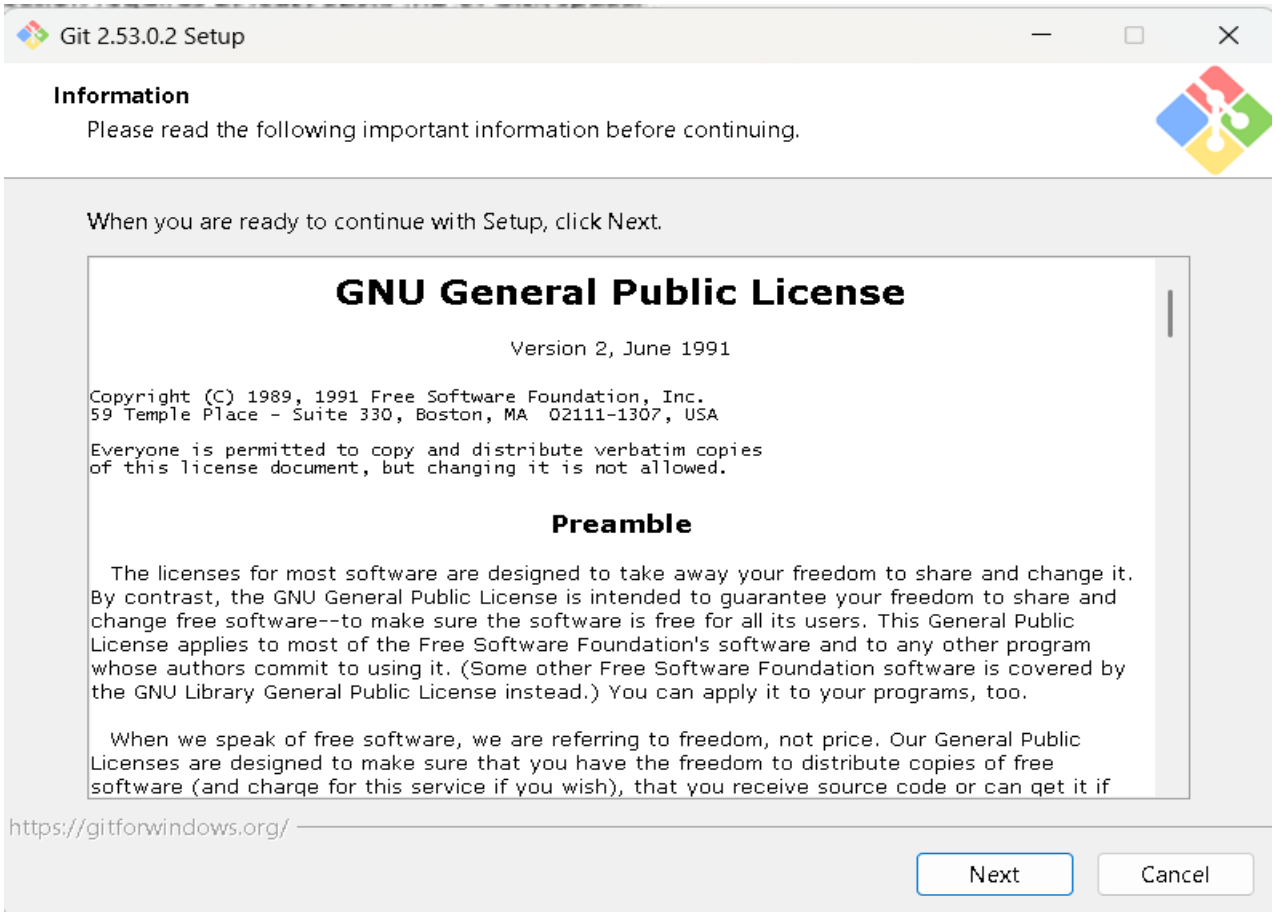
1. Installing Git
    a. Install Git on your local machine and verify the installation using git --version.
2. Configuring Git
    a. Set up your user name and email:
    b. git config --global user.name "Your Name"
    c. git config --global user.email "your.email@example.com"
3. Initializing a Repository
    a. Create a new directory and initialize it as a Git repository using git init.
    b. Explore the .git folder to understand its structure.
4. Open an account on www.github.com for an online repository which is available at anytime and anywhere.

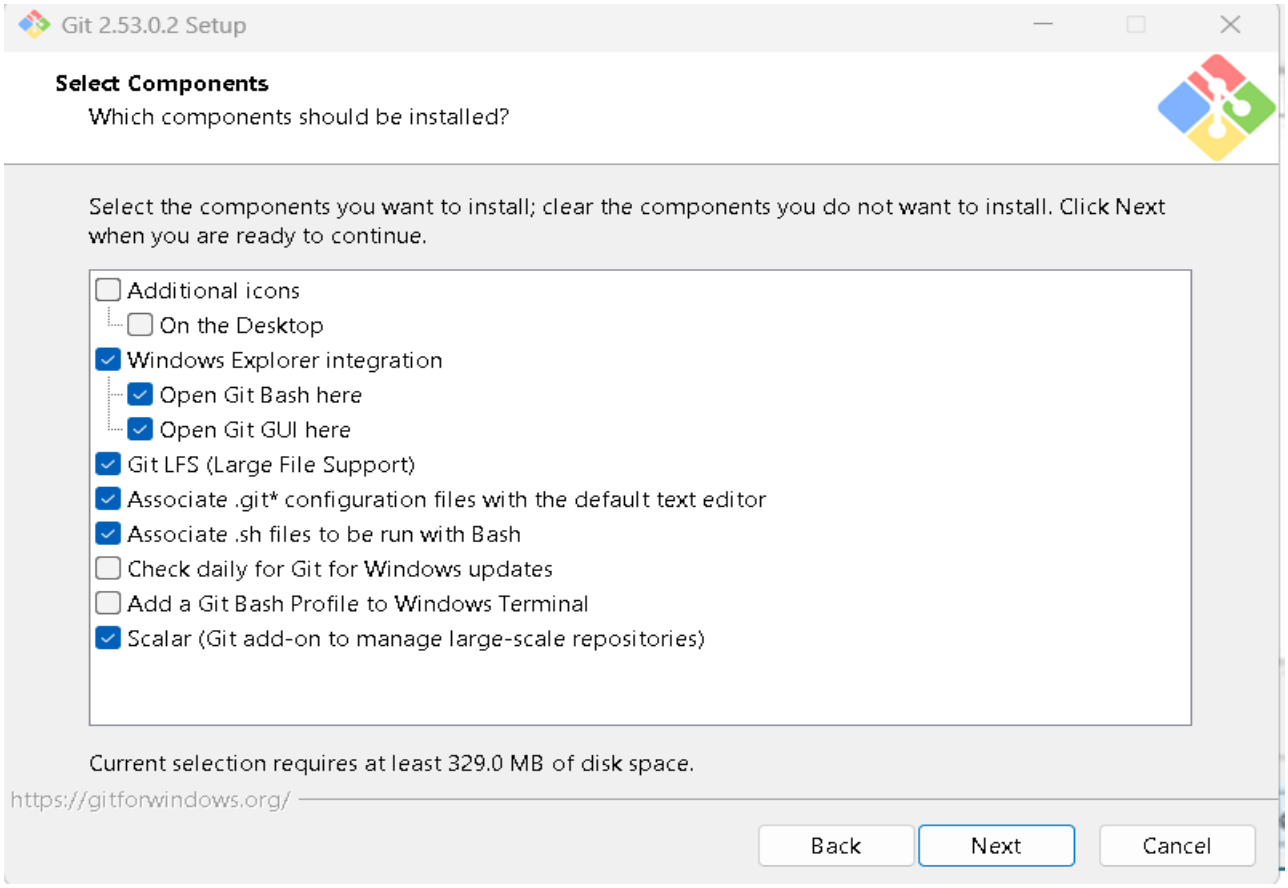
Exercise: -

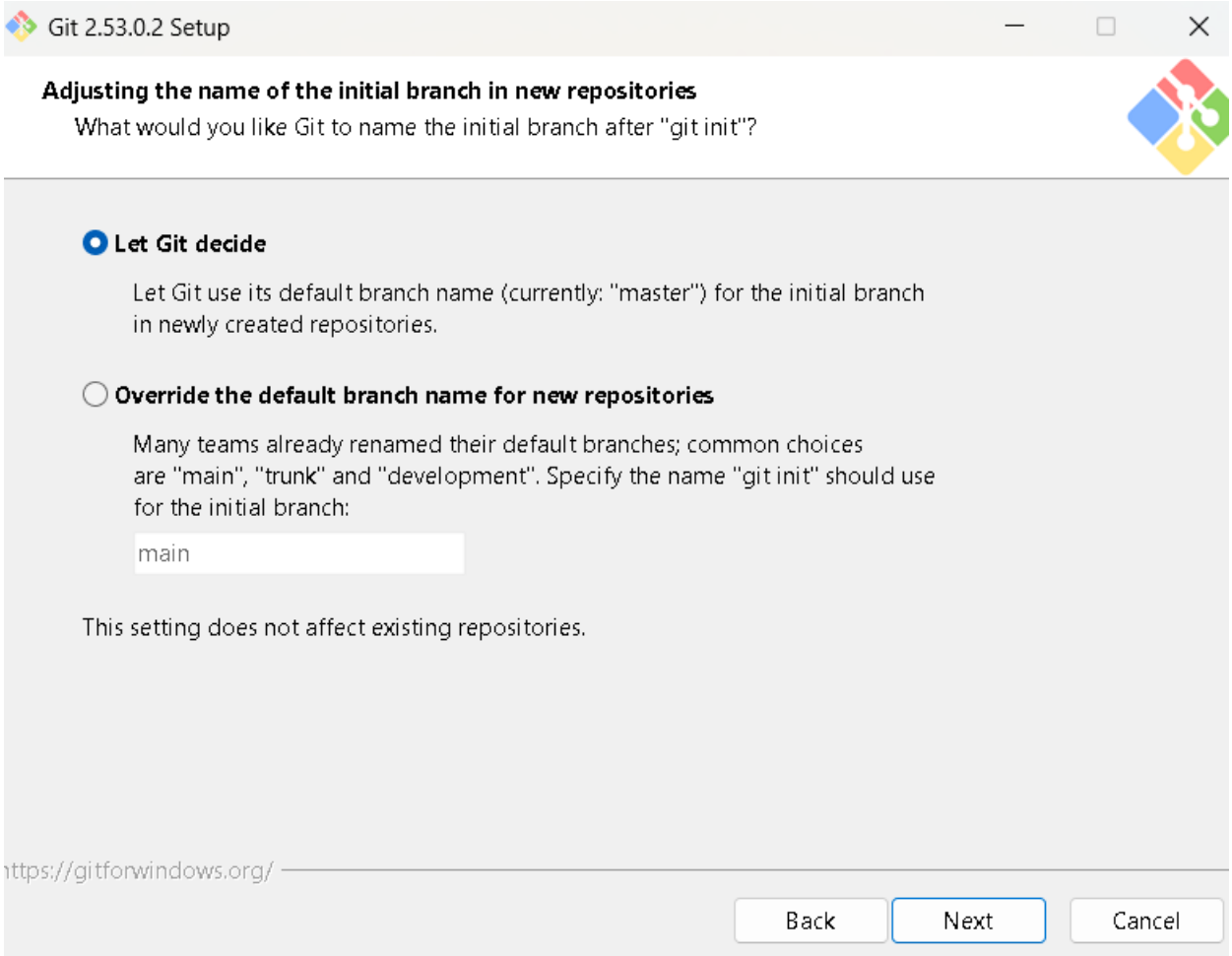
1. Team Setup: Form teams and assign roles (e.g., developer, tester, operations).
2. Git Installation: Download and Install git on your machine.
3. Configuration: Configure Git using your mail account and then verify it.
4. Initialization: Initialize or create a new Git repository and verify it.
5. Git Hub account: Sign in to www.github. com for an online repository which is available at anytime and anywhere.

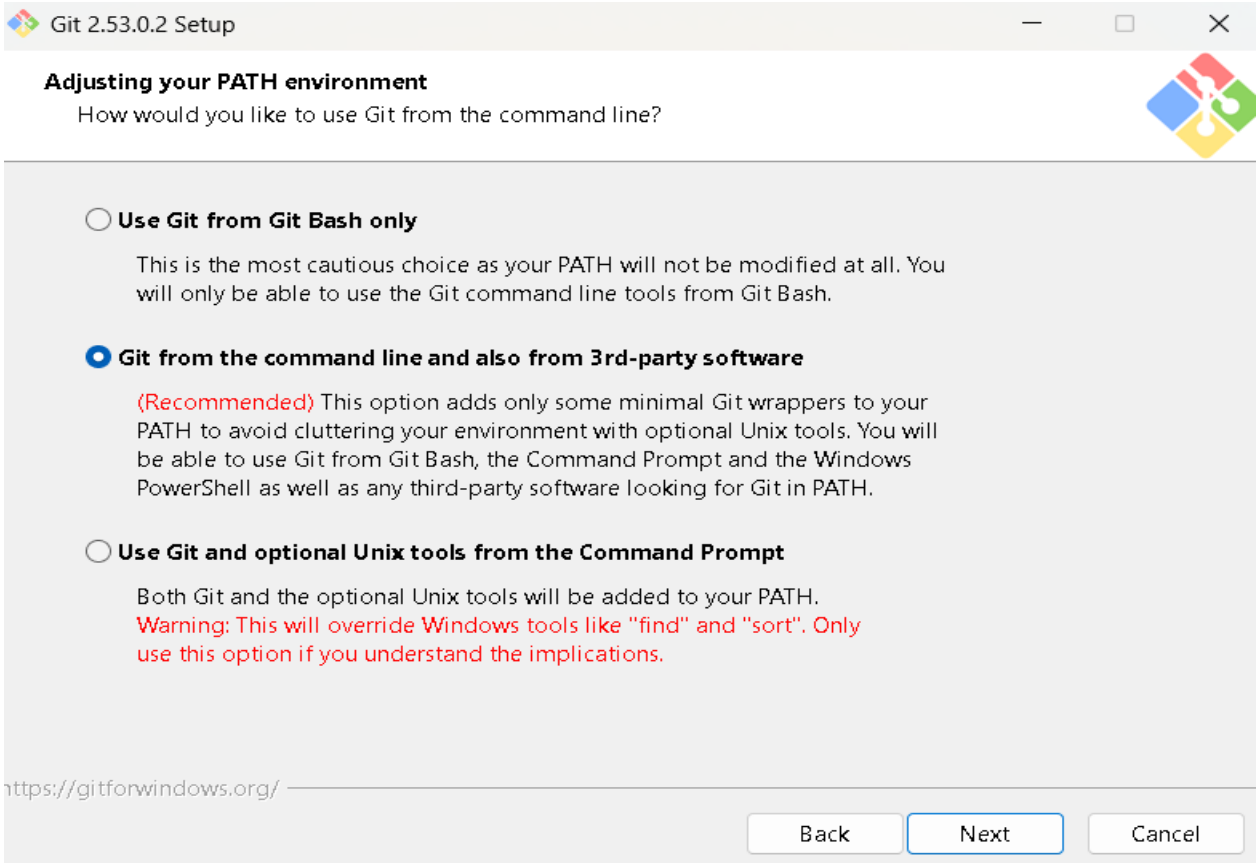
2. Git Installation: Download and Install git on your machine.

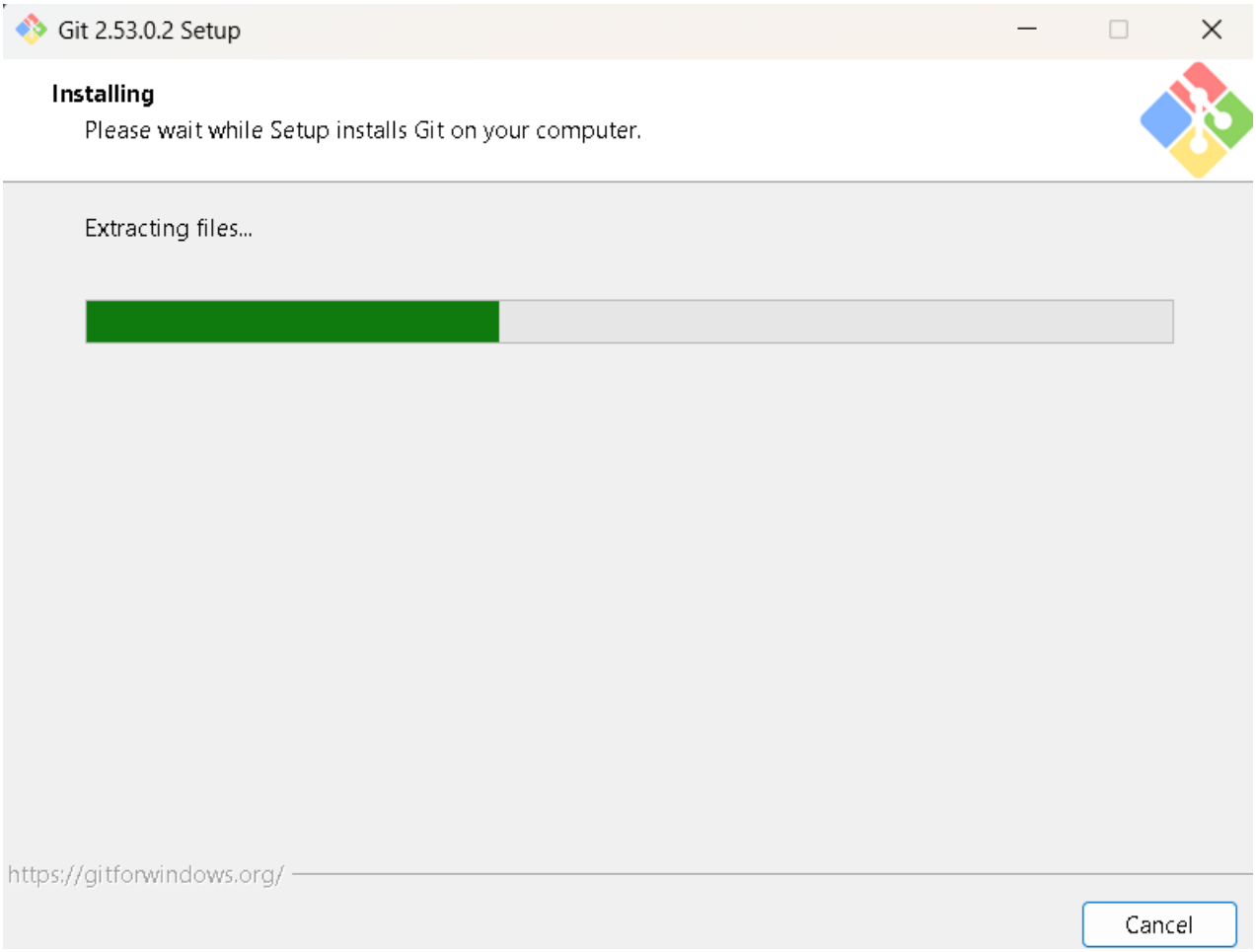


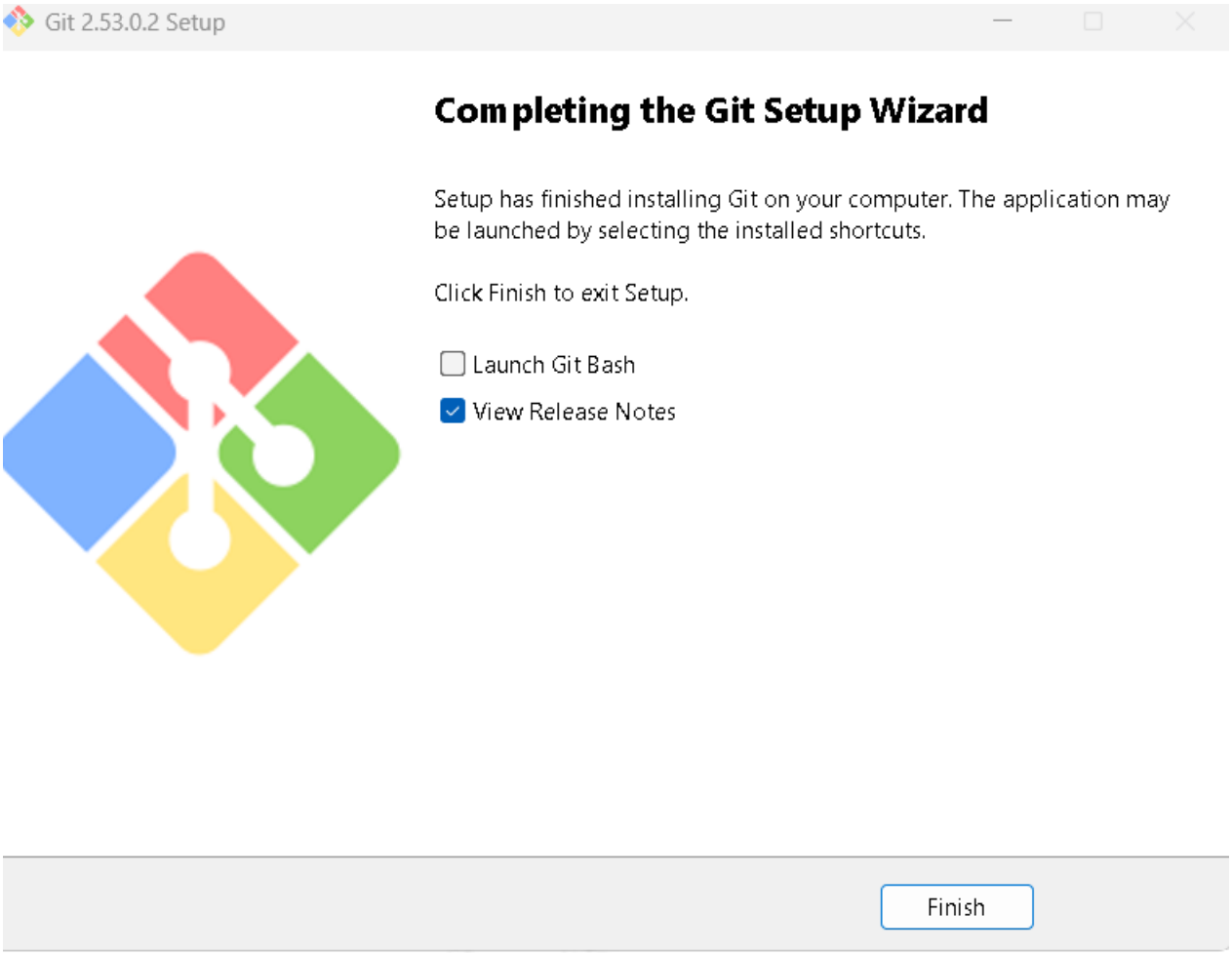


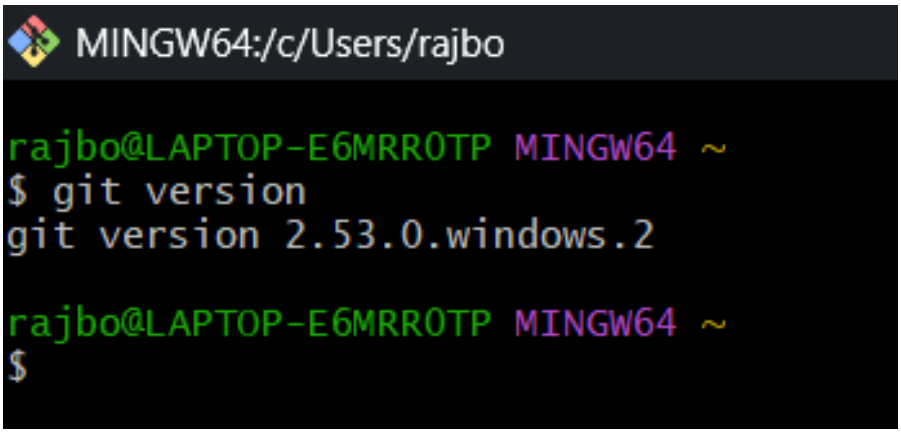

Git 2.53.0.2 Setup
Select Components
Which components should be installed?
Select the components you want to install; clear the components you do not want to install. Click Next when you are ready to continue.
Additional icons
On the Desktop
Windows Explorer integration
Open Git Bash here
Open Git GUI here
Git LFS (Large File Support)
Associate .git* configuration files with the default text editor
Associate .sh files to be run with Bash
Check daily for Git for Windows updates
Add a Git Bash Profile to Windows Terminal
Scalar (Git add-on to manage large-scale repositories)
Current selection requires at least 329.0 MB of disk space.
https://gitforwindows.org/
Back
Next
Cancel


Git 2.53.0.2 Setup
Adjusting the name of the initial branch in new repositories
What would you like Git to name the initial branch after "git init"?
Let Git decide
Let Git use its default branch name (currently: "master") for the initial branch in newly created repositories.
Override the default branch name for new repositories
Many teams already renamed their default branches; common choices are "main", "trunk" and "development". Specify the name "git init" should use for the initial branch:
main
This setting does not affect existing repositories.
https://gitforwindows.org/
Back
Next
Cancel

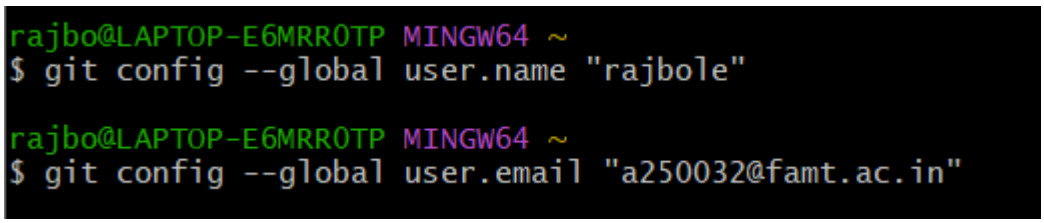

Git 2.53.0.2 Setup
Adjusting your PATH environment
How would you like to use Git from the command line?
Use Git from Git Bash only
This is the most cautious choice as your PATH will not be modified at all. You will only be able to use the Git command line tools from Git Bash.
Git from the command line and also from 3rd-party software
(Recommended) This option adds only some minimal Git wrappers to your PATH to avoid cluttering your environment with optional Unix tools. You will be able to use Git from Git Bash, the Command Prompt and the Windows PowerShell as well as any third-party software looking for Git in PATH.
Use Git and optional Unix tools from the Command Prompt
Both Git and the optional Unix tools will be added to your PATH.
Warning: This will override Windows tools like "find" and "sort". Only use this option if you understand the implications.
https://gitforwindows.org/
Back
Next
Cancel


Git 2.53.0.2 Setup
Installing
Please wait while Setup installs Git on your computer.
Extracting files...
https://gitforwindows.org/
Cancel





3. Configuration: Configure Git using your mail account and then verify it.



```
rajbo@LAPTOP-E6MRR0TP MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=rajbole
user.email=a250032@famt.ac.in
```

4. Initialization: Initialize or create a new Git repository and verify it.

```
rajbo@LAPTOP-E6MRR0TP MINGW64 ~
$ cd D:Sem-ll-46

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46
$ cd Devops

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops
$ git init
Initialized empty Git repository in D:/Sem-ll-46/Devops/.git/

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ ls -a
./  ../  .git/  Practicals/

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$
```

5. Git Hub account: Sign in to www.github. com for an online repository which is available at anytime and anywhere.

Sign up for GitHub

Email

a250032@famt.ac.in

Use a different Google account

Username*

RajBole-46

Your Country/Region*

India

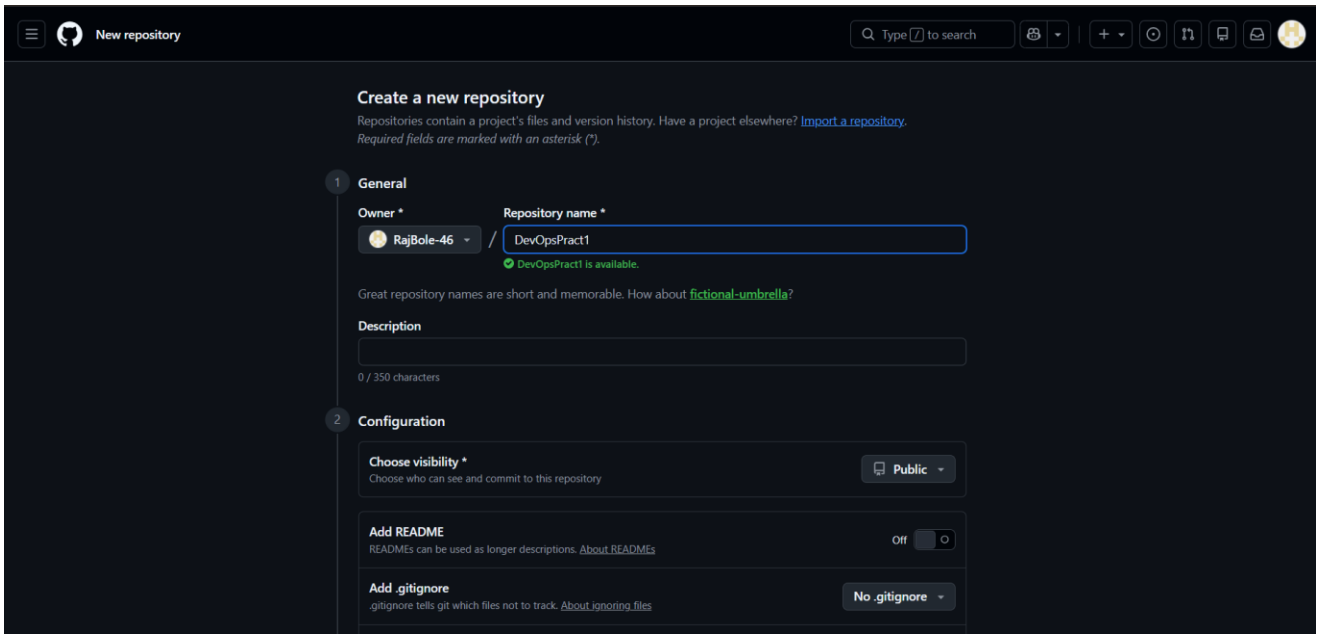
For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

Email preferences

☐ Receive occasional product updates and announcements

Create account ›

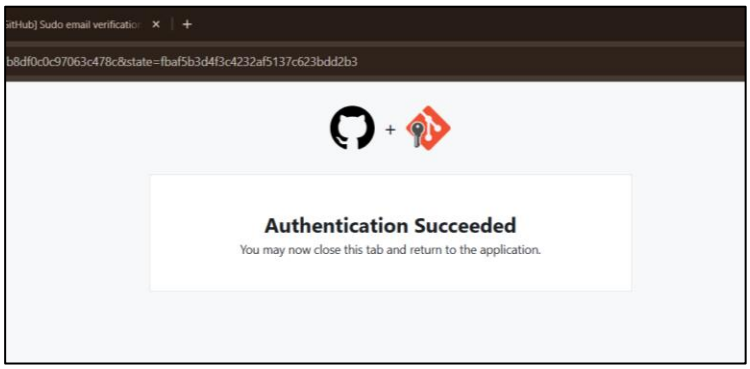
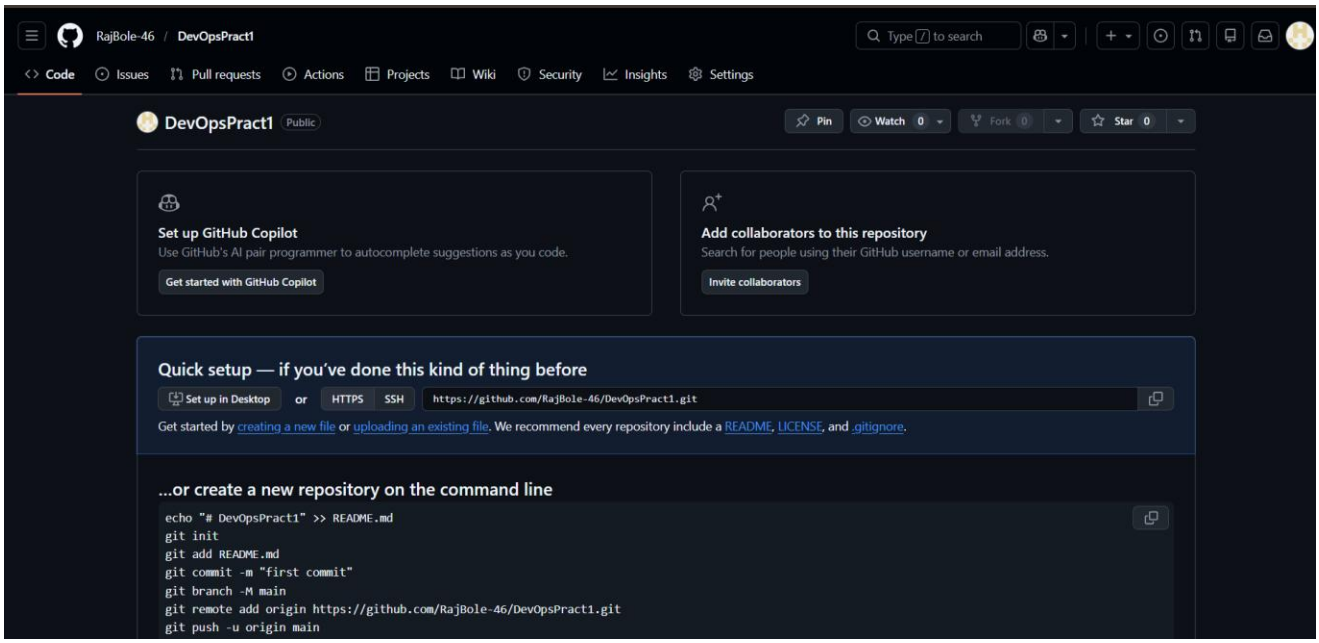
By creating an account, you agree to the Terms of Service. For more information about GitHub's privacy practices, see the GitHub Privacy Statement. We'll occasionally send you account-related emails.



```
rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git remote add origin https://github.com/RajBole-46/DevOpsPract1.git

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git remote --v
origin  https://github.com/RajBole-46/DevOpsPract1.git (fetch)
origin  https://github.com/RajBole-46/DevOpsPract1.git (push)

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ |
```

```
rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ echo "Hello World!" > demo.txt

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Demo.txt.txt
        Practicals/
        demo.txt

nothing added to commit but untracked files present (use "git add" to track)

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ |
```

```
rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git commit -m "This is first commit.."
[master (root-commit) 7c12bb3] This is first commit..
 1 file changed, 1 insertion(+)
 create mode 100644 demo.txt

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (master)
$ git branch -M main

rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (main)
$ git push -u origin main
info: please complete authentication in your browser...
```



```
rajbo@LAPTOP-E6MRR0TP MINGW64 /d/Sem-ll-46/Devops (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 225 bytes | 225.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/RajBole-46/DevOpsPract1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```